

Math 534H - Spring 2026 Lecture 23

Fourier Series and applications to the BIVP
for the wave and heat/diffusion equations on
the interval $[0, L]$. (roughly sections 5.1,
5.2, 5.3 and 5.4 of Chapter 5, Strauss' book).

I Basic Facts About Trigonometric Functions

Recall the following identities: for $a, b \in \mathbb{R}$

- $\cos a \cdot \cos b = \frac{1}{2} [\cos(a-b) + \cos(a+b)]$
- $\sin a \cdot \cos b = \frac{1}{2} [\sin(a-b) + \sin(a+b)]$

Definition: (Periodic functions on $\mathbb{R}$): A function ①

$\phi: \mathbb{R} \rightarrow \mathbb{R}$ (or $\phi: \mathbb{R} \rightarrow \mathbb{C}$) is said to be periodic if there exists a $p > 0$ such that $\phi(x+p) = \phi(x) \forall x \in \mathbb{R}$ (*)

Remark: If ϕ is periodic, the smallest such $p > 0$ for which (*) holds is called the period of ϕ .

Examples: i) $\sin x, x \in \mathbb{R}$ is periodic of period 2π .

ii) similarly $\cos x, x \in \mathbb{R}$ is periodic of period 2π .

iii) $\cos(2x), x \in \mathbb{R}$ is periodic of period π

iv) $\sin\left(\frac{x}{2}\right)$, $x \in \mathbb{R}$ is periodic of period 4π ⁽²⁾

v) $\tan x$ is not defined for all $x \in \mathbb{R}$ we must remove all the real numbers where $\cos x = 0$.
These is the set $\{\text{odd multiples of } \frac{\pi}{2}\}$. Then $\tan x$ is well defined for $x \in \mathbb{R} \setminus \{\text{odd multiples of } \frac{\pi}{2}\}$ and on this domain, $\tan x$ is periodic of period π . Note here we need to adjust the domain

Remarks ① If ϕ has period p , then:

$$\phi(x + np) = \phi(x) \quad \forall n \in \mathbb{Z}, x \in \mathbb{R}$$

② Given 2 periodic functions f of period P_1 ③ and g of period P_2 ($P_1, P_2 > 0$) we have:

① If $P_1 = m\pi$ and $P_2 = n\pi$ ($m, n \in \mathbb{N}$)

then $f \pm g$ is periodic of period $\max\{P_1, P_2\}$.

② If ① does not hold we have that $f \pm g$ is periodic **ONLY IF** ! $\frac{P_1}{P_2} \in \mathbb{Q}$

(that is only if $\frac{P_1}{P_2}$ is a rational number).

In this case, the period P of $f \pm g$ is the

lowest common multiple of P_1 and P_2 . (4)

Example (i) $\sin x$ has period 2π
 $\tan x$ has period π } $\Rightarrow$ by (a)

$\sin x + \tan x$ has period 2π .

(ii) $\cos(\pi x)$ is periodic of period 2.

$\cos(\sqrt{2}x)$ is periodic of period $\sqrt{2} \cdot \pi$

But since $\frac{2}{\sqrt{2} \cdot \pi} = \frac{\sqrt{2}}{\pi} \notin \mathbb{Q}$ (not rational)

$\cos(\pi x) + \cos(\sqrt{2}x)$ is NOT periodic

(3) If $\phi: \mathbb{R} \rightarrow \mathbb{R}$ is periodic of period $P > 0$.
Assume that ϕ is integrable.

Then we have that $\int_a^{a+P} \phi(x) dx$ is independent ⁽⁵⁾
of a , for any $a \in \mathbb{R}$.

In particular, we have:

$$\int_a^{a+P} \phi(x) dx = \int_0^P \phi(x) dx \quad \forall a \in \mathbb{R}$$

(this is not so straightforward to prove. You'll have a HWK problem about it).

④ A periodic function of period $P > 0$ is fully specified if we give its values on

the interval $(0, P)$ or on any other interval I of length P (for example; $(-\frac{P}{2}, \frac{P}{2})$ or $(\sqrt{2}, \sqrt{2} + P)$, etc).

Suppose now that we are given a function on an interval I of length $2l$. Then such a function CAN BE EXTENDED to all of $\mathbb{R}$ as a PERIODIC FUNCTION of period $2l$ in ONLY ONE WAY: More precisely suppose that $\phi: (-l, l) \rightarrow \mathbb{R}$ is given. Then its

periodic extension

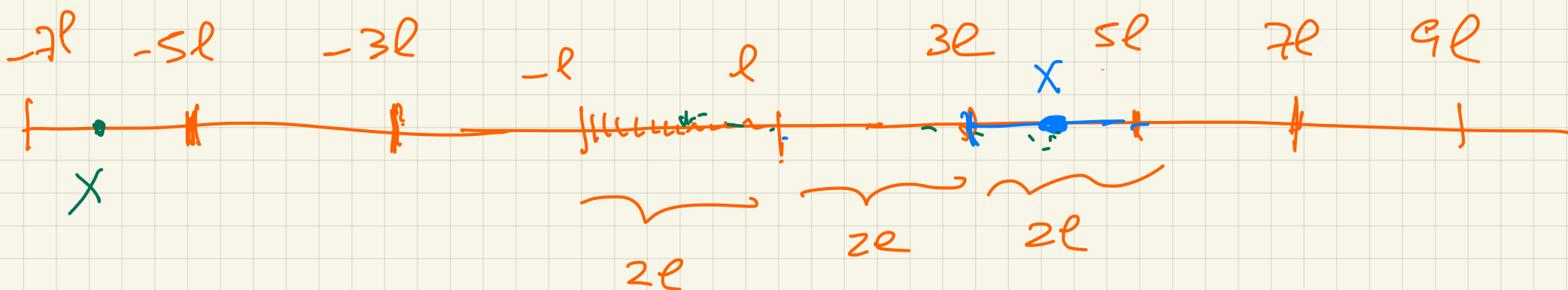
defined by : $\phi_{\text{per}} : \mathbb{R} \rightarrow \mathbb{R}$ is

$$\phi_{\text{per}}(x) := \phi(x - 2\ell m) \text{ for}$$

$$-\ell + 2\ell m < x < \ell + 2\ell m, \forall m \in \mathbb{Z}$$

This is called the periodization of ϕ with period 2ℓ .

IMPORTANT REMARK : Note that this definition of $\phi_{\text{per}}(x)$, $x \in \mathbb{R}$ DOES NOT specify the values



$$\phi_{\text{per}}(x) = \phi(x - (2l) \cdot m)$$

$$-l + 4l < x < l + 4l$$

$$-l + 2(2l) < x < l + 2(2l)$$

$$+2 = m$$

$$-l - 3(2l) < x < l - 3(2l)$$

$$-3 = m$$

$$-7l < x < -5l$$

of ϕ_{per} at the endpoints $x = -l + 2m\ell$
and $x = l + 2m\ell$, $m \in \mathbb{Z}$.

In fact, ϕ_{per} may be discontinuous at
these endpoints unless

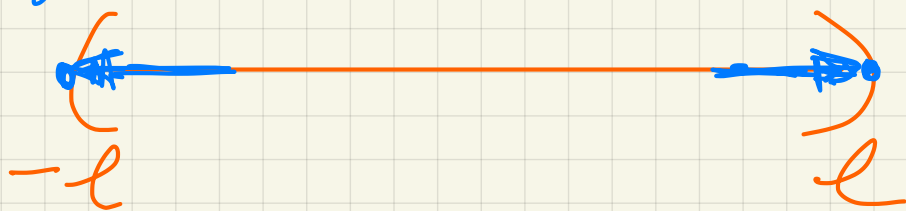
$$\phi(l^-) = \phi(-l^+)$$

In other words unless

$$\lim_{x \rightarrow l^-} \phi(x) = \lim_{x \rightarrow -l^+} \phi(x).$$

$(x < l)$

$(x > -l)$ (unless the side
limits at the
endpoints are
equal).



We finally recall the following (PROVE THEM!)⁽⁹⁾

$$\textcircled{1} \int_0^l \sin\left(\frac{n\pi x}{l}\right) \sin\left(\frac{m\pi x}{l}\right) dx = \begin{cases} 0 & n \neq m \\ \frac{l}{2} & n = m \quad (\neq 0). \end{cases}$$

$$\textcircled{2} \int_0^l \cos\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{m\pi x}{l}\right) dx = \begin{cases} 0 & \text{if } n \neq m \\ \frac{l}{2} & \text{if } n = m \quad (\neq 0) \end{cases}$$

Note: In case $\textcircled{2}$ if $n = m = 0$ we get on the left hand side $\int_0^l 1 dx$ which $= l$.

In case $\textcircled{1}$ if $n = m = 0$ the left and right hand side

are 0).

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$$\textcircled{3} \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) \sin\left(\frac{m\pi x}{l}\right) dx = 0 \quad \forall n, m \in \mathbb{Z}$$

$$\textcircled{4} \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{m\pi x}{l}\right) dx = 0 \quad \underline{\text{for } n \neq m}$$

$$\textcircled{5} \int_{-l}^l \sin\left(\frac{n\pi x}{l}\right) \cdot \sin\left(\frac{m\pi x}{l}\right) dx = 0 \quad \underline{\text{for } n \neq m}$$

$$\textcircled{6} \int_{-l}^l 1 \cdot \cos\left(\frac{n\pi x}{l}\right) dx = 0 = \int_{-l}^l 1 \cdot \sin\left(\frac{n\pi x}{l}\right) dx \quad \textcircled{11}$$

$$\forall n \in \mathbb{Z}.$$

$$\textcircled{7} \int_{-l}^l \cos^2\left(\frac{n\pi x}{l}\right) dx = l = \int_{-l}^l \sin^2\left(\frac{n\pi x}{l}\right) dx$$

$$\forall n \in \mathbb{Z}.$$

NOTE: The equalities $\textcircled{1}$ through $\textcircled{6}$ give us important orthogonality relationships of the trigonometric system $\left\{ \sin\left(\frac{n\pi x}{l}\right), \cos\left(\frac{n\pi x}{l}\right) \right\}_{n \in \mathbb{Z}}$.

We now come back to the Boundary Initial Value Problem for the wave equation and for the heat/diffusion equation on an interval I which we relabel by $(0, \ell) = I$.

- We start with the wave equation with zero Dirichlet boundary conditions. Recall that in this case we were led to the question of whether/when the initial position $\phi(x)$ and the initial velocity $\psi(x)$ could be represented

by SINE SUMS/SERIES. Namely:

(13)

$$(*) \quad \phi(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{l}\right) \quad 0 < x < l$$

$$(**) \quad \psi(x) = \sum_{n=1}^{\infty} B_n \frac{\pi n c}{l} \sin\left(\frac{n\pi x}{l}\right) \quad 0 < x < l$$

($c > 0$ speed of the wave)

for suitable coefficients A_n, B_n .

Thus to answer this question we need to

(a) Find A_n, B_n associated to ϕ and ψ respectively.

(b) Understand in what sense each series converges to ϕ and ψ . That is in what sense the series converge and in what sense the equalities $\stackrel{?}{=}$ hold.

Remark: Similarly for the heat/diffusion equation with zero Dirichlet BC we were led to a similar sine sum/series as (*) in the previous page but now $\phi(x)$ was the initial temperature or initial concentration.

• We focus NEXT in answering question (a).